

PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KILOLO DISTRICT COUNCIL
UHAMBINGETO SECONDARY SCHOOL
FORM THREE TOPICAL EXAMINATION ON THURSDAY 23TH APRIL 2026.

041

BASIC MATHEMATICS

TIME: 3HRS

wazaelimu.com

INSTRUCTIONS

- *This paper consists of section A and B*
- *Answer all questions on both sections*
- *Remember to write your name on every page of your answer sheets.*
- *Non programmable calculator may be used in the examination room*

SECTION A (60 Marks)

Answer all questions

1. (a) Find the next two terms of each of the following sequences
 - i). $\frac{1}{2}, 2/3, 3/4, \dots$
 - ii). 1, 2, 6, 24, 120, ...(b) If the n^{th} of a sequence is given by $2n+1$, find the tenth term.
2. (a) Find the general term of the sequence 3, 7, 11, 15, 19, ...
(b) the general term of a certain sequence is $\frac{1}{2+(-1)^n}$, Find the first four terms. Is the sequence increasing or decreasing? Give reasons.
3. Consider the following series $a+(a+d)+(a+2d)+(a+3d)+\dots$, Find the:
 - i). Difference between the second and the first term
 - ii). Difference between the third term and the second term
 - iii). Difference between the fifth term and the fourth term
 - iv). Expression of the general term
4. A Lecture Hall has 20 rows of seats, with 100 seats in the back row. Each row has 2 fewer number of seats than the row immediately behind it. How many seats are there in the Lecture Hall?
5. The fourth and fifth terms of an Arithmetic progression are 47 and 52 respectively. Find,
 - i). The common difference
 - ii). The first term
 - iii). The thirteenth term.
6. (a) Find the n^{th} of an Arithmetic Progression whose first term is $x+2$ and the common difference is 3.
(b) The n^{th} term of an Arithmetic Progression $2.3+4.2+6.1+\dots$ is 36.5. Find the value of n .
7. The numbers 24, m , y and 72 are such that m is the arithmetic mean of 24 and y , while y is the arithmetic mean of m and 72. Find the values of m and y .

8. The first term of an Arithmetic Progression is 6 and the common difference is 1.5. If the sum of the first n terms is 90, find the number of terms of the progression.
9. (a) If the fourth term of a Geometric progression is 9 and the sixth term is 81, find:
- The common ratio
 - The first term
- (b) The second and fifth terms of a geometric progression are 24 and 5184 respectively. Find;
- The first term
 - The common ratio
 - The seventh term
10. (a) Find the sum of the first twenty terms of the geometric series $4+8+16+\dots$
- (b) If the n^{th} term of a geometric progression is given by $G_n=2^n$, find the sum of the first five terms.

SECTION B (40 Marks)

In this section answer all questions

11. (a) The geometric Mean of two numbers 4 and P is 24. Find the value of P
- (b) If the sum of the first n terms is 3^n-1 , show that the sequence is a geometric progression.
12. (a) The seventh term of Geometric Progression (G.P) is eight times the fourth term and fifth term is 48. Find the first term and the Common ratio.
- (b) The n^{th} term of an A.P is $12-4n$. Find the first term and the Common difference
13. The third, fourth and eighth terms of Arithmetic Progression (A.P) forms the first three consecutive terms of the Geometric Progression (G.P), If the sum of the first 10 terms of the Arithmetic Progression is 85. Find:
- The first term of an A.P
 - The Common ratio of G.P
 - The sum of the first 7 terms of G.P
14. (a) Mr.Ndelemo deposited 10,000 Tshs in the NBC bank at the rate of 8% Compounded annually for 3 years. Calculate the amount deposited after 3 years.
- (b) Subwenke invested a certain amount of money in CRDB bank whose interest rate was 12% compounded annually. After three years, he got 13,310,000 Shillings. How much did he invest initially?

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Prepared by,

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